torch.nn.functional.bilinear#

https://pytorch.org/docs/stable/generated/torch.nn.functional.bilinear.html

Description:

Applies a bilinear transformation to the incoming data: y=x1TAx2+by = x_1^T A x_2 + by=x1TAx2+b input1: (N,∗,Hin1)(N, *, H_{in1})(N,∗,Hin1) where Hin1=in1_featuresH_{in1}=\text{in1_features}Hin1=in1_features and ∗*∗ means any number of additional dimensions. All but the last dimension of the inputs should be the same. input2: (N,∗,Hin2)(N, *, H_{in2})(N,∗,Hin2) where Hin2=in2_featuresH_{in2}=\text{in2_features}Hin2=in2_features weight: (out_features,in1_features,in2_features)(\text{out_features}, \text{in1_features}, \text{in2_features})(out_features,in1_features,in2_features)

Function Signature

```
torch.nn.functional.bilinear(input1, input2, weight, bias=None) → Tensor#
```

Detailed Documentation

Documentation

Applies a bilinear transformation to the incoming data: y=x1TAx2+by = x_1^T A x_2 + by=x1TAx2+b

input1: (N,∗,Hin1)(N, *, H_{in1})(N,∗,Hin1) where Hin1=in1_featuresH_{in1}=\text{in1_features}Hin1=in1_features and ∗*∗ means any number of additional dimensions. All but the last dimension of the inputs should be the same.

input2: (N,∗,Hin2)(N, *, H_{in2})(N,∗,Hin2) where Hin2=in2_featuresH_{in2}=\text{in2_features}Hin2=in2_features

weight: (out_features,in1_features,in2_features)(\text{out_features}, \text{in1_features}, \text{in2_features})(out_features,in1_features,in2_features)

bias: (out_features)(\text{out_features})(out_features)

output: (N,∗,Hout)(N, *, H_{out})(N,∗,Hout) where Hout=out_featuresH_{out}=\text{out_features}Hout=out_features and all but the last dimension are the same shape as the input.